

CE 310

Probability and Bayes' Theorem

Week 3 · Session 1 · Classical Probability, Conditional Probability, and Bayes

University of Arizona · Dr. Wooyoung Jung · Fall 2026

Part 1

Why Conditioning Changes Everything

The Night Before the Launch

90 seconds · you are in the teleconference

It is the evening of **January 27, 1986**. Tomorrow's shuttle launch is under review because the forecast is cold. Here is the summary in front of you:

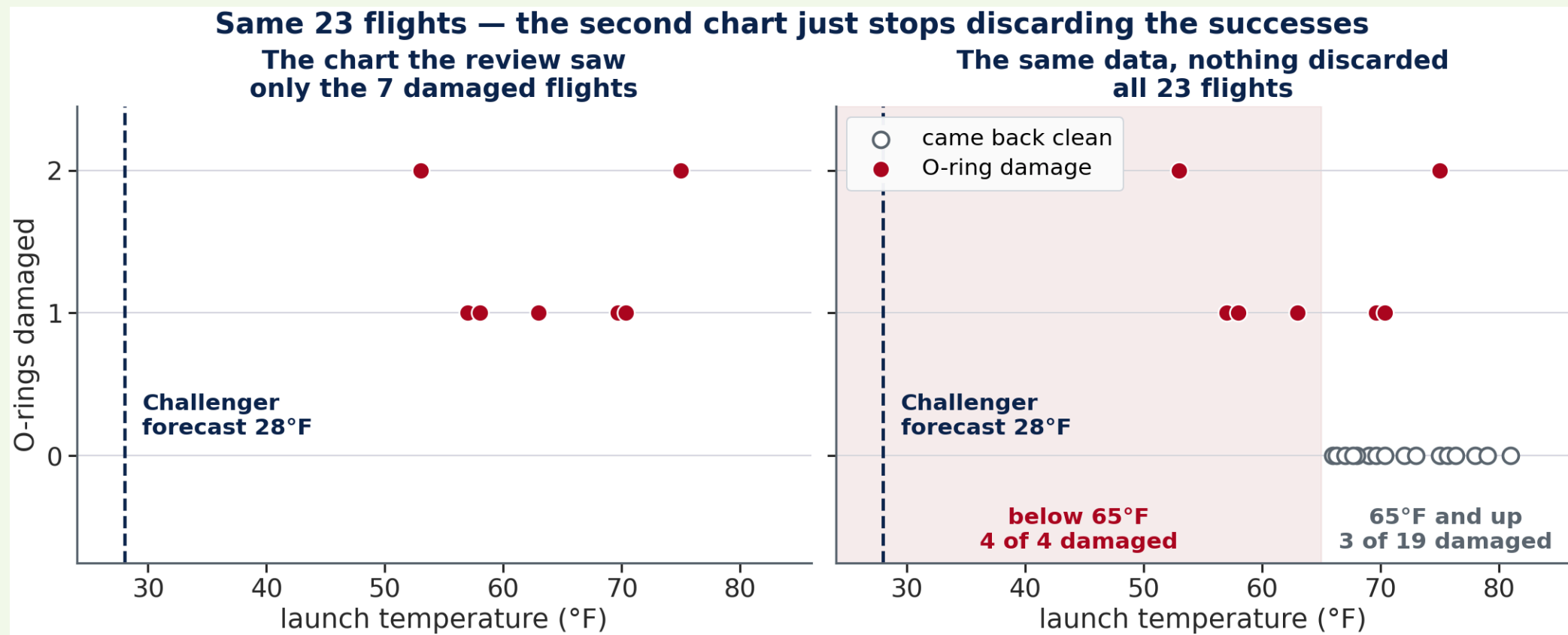
- **23 previous flights** with recovered boosters. **7 showed O-ring damage** — an overall damage rate of about **30%**
- **Every one of those 23 flights came home safely**, damaged O-rings included
- The **coldest launch in program history was 53°F**
- Tomorrow's forecast puts the joints near **28°F** — 25 degrees below anything ever flown

Two questions. Commit to both before the next slide:

1. From this summary alone, is there a statistical basis to stop the launch?
2. What is the *one* thing you would ask to see?



The Question Was: Damage Given What?

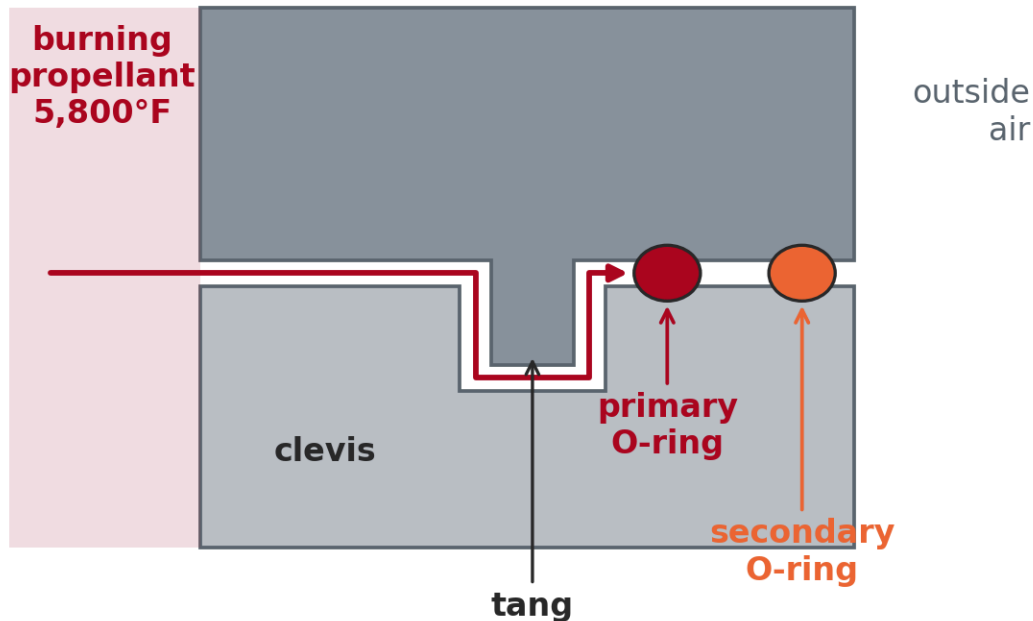


What an O-ring Is — and Why 30 Degrees Mattered

The O-ring is the only thing between the fire and the outside

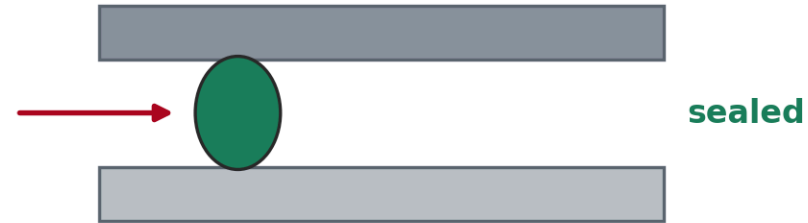
A booster is stacked from segments. Every joint is a seam plugged by two rubber rings.

The ring has to move faster than the gap opens.



At ignition the case balloons and the gap opens.
Warm rubber follows it about 5× faster than cold.

75°F — rubber springs back, fills the gap



28°F — rubber is stiff, lags behind



The Launch That Should Not Have Happened

January 28, 1986 · Space Shuttle Challenger · joints at 28°F

- O-ring seals failed in the cold → shuttle broke apart 73 seconds after launch
- Seven crew members died
- Engineers had raised concerns the night before



73 seconds after launch. NASA/KSC — public domain.

Nobody was missing data. Every flight in the chart you just read had already flown. The failure was that the risk was summarized as an **overall damage rate**, and no one put that rate next to a thermometer.

⚠ P(damage) = 30% across all 23 flights. $P(\text{damage} \mid T < 65^\circ\text{F}) = 100\%$ — four launches out of four. The marginal probability was not wrong. It was an average over two populations, and it described the one that mattered least.

Three CE/ArcE Decisions — All Require Conditioning

Design Decision	Wrong approach	What you actually need
Size HVAC cooling system	Annual average cooling (19 kWh/hr)	$P(\text{Demand} > \text{threshold} \mid \text{summer business hours})$
Size backup generator	Marginal $P(\text{Electricity} > 100) = 39\%$	$P(\text{Electricity} > 100 \mid \text{business hours}) \approx \mathbf{96\%}$
Schedule preventive maintenance	Overall fault rate	$P(\text{Fault} \mid \text{sensor reading anomalous})$

💡 The third row is different: you know what the sensor says, and you want to know what that means about the equipment. That requires **reversing** the conditional — computing $P(\text{fault given alert})$, not $P(\text{alert given fault})$. That is Bayes' theorem.

Where We Left Off — A Marginal Probability

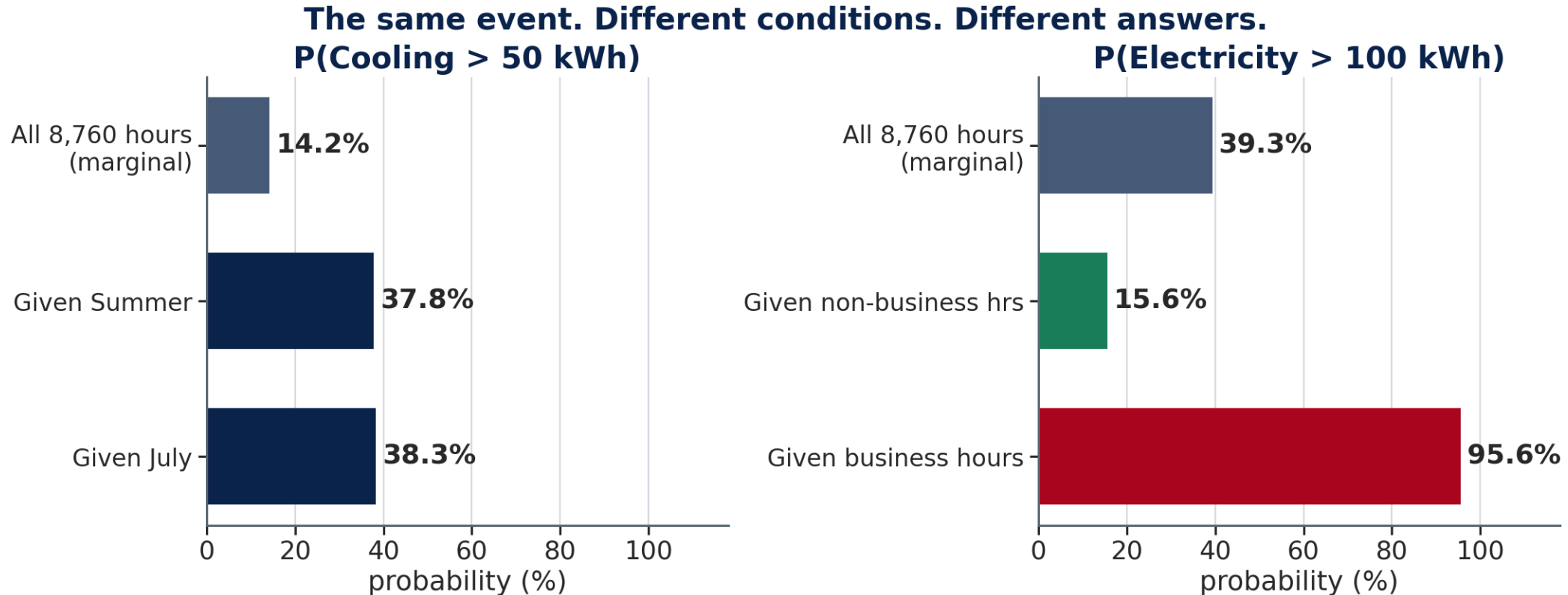
In the Tucson 2023 dataset, you counted 900 hours where $T > 95^\circ\text{F}$. Using that count:

$$P(T > 95^\circ F) = \frac{900 \text{ hours}}{8,760 \text{ hours}} = \mathbf{10.3\%}$$

This is a **marginal probability** — no conditioning on month, time of day, or occupancy.

Week 3 takes the next step: instead of $P(\text{event})$, we ask $P(\text{event} \textbf{ given } \text{some context})$. And then we ask what happens when you need to flip that conditional around.

Conditioning Changes the Answer — Not a Little



Part 2

Classical Probability

What Is Probability?

 **Probability:** A number between 0 and 1 that quantifies how likely an event is to occur. $P = 0$ means impossible; $P = 1$ means certain.

Three interpretations — all valid in engineering:

Frequency

$P(A)$ = fraction of times A occurs over many repeated trials

"This chiller faults in 5% of operating hours"

Classical

$P(A)$ = favorable outcomes / total equally likely outcomes

" $P(\text{Monday}) = 1/7$ for a randomly chosen weekday"

Subjective

$P(A)$ = degree of belief, updated with evidence

"I believe there is a 70% chance this component needs replacement"

Sample Space, Events, and the Three Rules

Sample space Ω : the set of all possible outcomes

Rule	Formula	Plain meaning
Complement	$P(A^c) = 1 - P(A)$	If $P(\text{rain}) = 0.3$, then $P(\text{no rain}) = 0.7$
Addition	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	Probability of A or B (or both)
Mutually exclusive	$P(A \cup B) = P(A) + P(B)$ if $A \cap B = \emptyset$	A and B cannot both occur

 **Key distinction:** The addition rule subtracts the overlap $P(A \cap B)$ to avoid double-counting. When events are mutually exclusive the overlap is zero — so the simpler formula applies.

Your Turn — Addition Rule

A building has two independently monitored systems:

- $P(\text{HVAC fault in any given week}) = \mathbf{0.05}$
- $P(\text{Electrical fault in any given week}) = \mathbf{0.03}$

What is $P(\text{at least one fault occurs this week})$?

Think before calculating. A colleague says "just add them — $5\% + 3\% = 8\%$." Is that right?

✓ Addition Rule — Answer

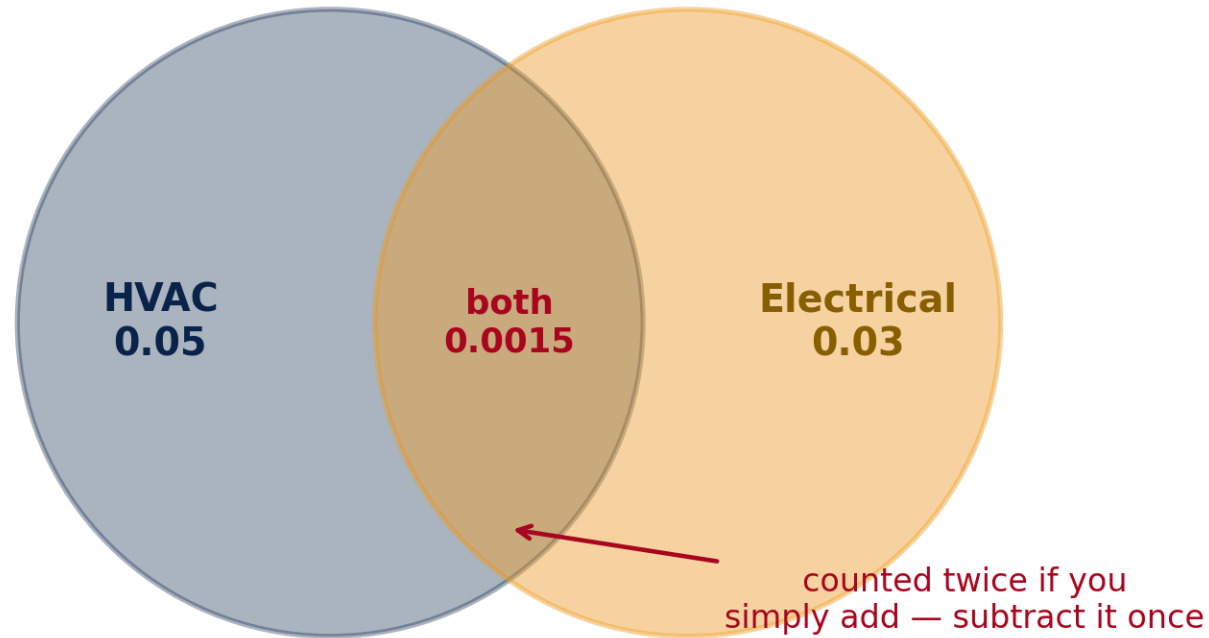
$$\begin{aligned} P(\text{some fault}) &= P(\text{HVAC}) + P(\text{Elec}) - P(\text{HVAC} \cap \text{Elec}) \\ &= 0.05 + 0.03 - (0.05 \times 0.03) = 0.08 - 0.0015 = \mathbf{0.0785} \end{aligned}$$

Not 0.08 — the overlap (both systems failing in the same week) is double-counted if you simply add, and must be subtracted.

💡 The gap is small here (0.0015). In reliability systems with many components, uncorrected overlaps compound — and the error grows large enough to matter for safety calculations.

Why You Subtract the Overlap

$$P(A \cup B) = 0.05 + 0.03 - 0.0015 = 0.0785, \text{ not } 0.08$$



Quick Check — Which Rule Applies?

90 seconds · three questions

A building automation system monitors two subsystems independently:

- $P(\text{HVAC fault in any given week}) = 0.10$
- $P(\text{Lighting fault in any given week}) = 0.04$

Question A: What is $P(\text{no HVAC fault this week})$?

Question B: What is $P(\text{HVAC fault OR lighting fault this week})$?

Question C: A colleague says "just add them — $10\% + 4\% = 14\%$." When is that answer exactly right, and when is it wrong?

Quick Check — Answer

A: $P(\text{no HVAC fault}) = 1 - 0.10 = \mathbf{0.90}$

B: $P(\text{HVAC OR lighting fault}) = 0.10 + 0.04 - (0.10 \times 0.04) = \mathbf{0.136}$

C: Adding directly gives the right answer **only when the events are mutually exclusive** — when they cannot both happen at once. Since both faults can occur in the same week, there is an overlap (both fault simultaneously with probability $0.10 \times 0.04 = 0.004$) that must be subtracted.

 **Mutually exclusive vs. independent:** Mutually exclusive means they cannot both occur — $P(A \cap B) = 0$. Independent means knowing one happened tells you nothing about the other — $P(A \cap B) = P(A) \times P(B)$.

Part 3

Conditional Probability

What Does "Given" Mean?

 **Conditional probability:** $P(A \mid B)$ is the probability of A given that B has already occurred — or given that we restrict our view to the subset of outcomes where B is true.

Formula:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{\text{Count}(A \text{ AND } B)}{\text{Count}(B)}$$

Rearranged — the multiplication rule: $P(A \cap B) = P(A \mid B) \cdot P(B)$ — a conditional times the probability of its condition gives the joint. A fraction *of* a fraction. This one holds **always**, independent or not, and every Bayes calculation in Part 4 is built on it.

Intuition: You started with the full sample space Ω . Conditioning on B shrinks your sample space to only the outcomes where B occurred. You then ask: within that smaller world, how often does A happen?

Key implication: $P(A \mid B) \neq P(A)$ whenever A and B are **associated** — when knowing B changes what you expect about A.

"Given" Means: Throw the Other Rows Away

Conditioning shrinks the denominator — that is the whole idea

Step 1 — all 8,760 hours



July • 744 hrs

Step 2 — July becomes the whole world



$$P(\text{Cooling} > 50 \mid \text{July}) = 285 / 744 = 38.3\%$$

Predict — Conditioning on Month

In this week's in-class exercise you will compute $P(\text{Cooling} > 50 \text{ kWh} \mid \text{Month})$ for each month of 2023.

The formula:

$$P(\text{Cooling} > 50 \mid \text{July}) = \frac{\text{July hours with Cooling} > 50}{\text{all July hours}} = \frac{?}{744}$$

What this does: the denominator is no longer 8,760 — it is 744, the hours July actually has. You are asking what fraction of *those* hours had high cooling demand.

Overall probability (all 8,760 hours): $P(\text{Cooling} > 50 \text{ kWh}) = 14.2\%$

Your prediction: what is $P(\text{Cooling} > 50 \text{ kWh} \mid \text{Month} = \text{July})$? Higher? Lower? By how much?

✓ **$P(\text{Cooling} > 50 \text{ kWh} \mid \text{July}) = 38.31\%$**

Marginal (all 8,760 hours):

$P(\text{Cooling} > 50 \text{ kWh}) = 14.2\%$

Conditional (July only):

$P(\text{Cooling} > 50 \text{ kWh} \mid \text{July}) = 38.31\%$

Conditioning multiplies the probability by **2.7×**
— nearly three times higher in July alone.

💡 The 14.2% is the right answer to the wrong question. For peak-load HVAC sizing, the right question is July (or July during business hours) — not the annual average. Conditional probability makes that gap visible.

A Second Domain — Storm Drainage Design

Conditioning is not an HVAC-only idea. A drainage engineer reviews 10 years of Tucson monsoon storm records (142 total storm events) while sizing a culvert.

Marginal probability (any storm):

$$P(\text{Culvert overtops}) = \frac{9}{142} = \mathbf{6.3\%}$$

Conditioning on storm intensity: of the 142 storms, 11 were classified as ≥ 25 -year design intensity. The culvert overtopped in 7 of those 11.

$$P(\text{Culvert overtops} \mid \text{storm} \geq 25\text{-yr}) = \frac{7}{11} = \mathbf{63.6\%}$$

💡 A culvert that "only overtops 6% of the time" overtops **10× more often** during the storms that actually stress it. Sizing to the marginal rate means the culvert fails nearly two-thirds of the time it matters most — the same mistake as the HVAC and generator examples, in a different discipline.

Quick Check — Conditioning in a New Domain

90 seconds

A transportation department reviews 3 years of crash reports at a signalized intersection (1,200 total observed days).

- Crashes occurred on 54 of the 1,200 days $\rightarrow$ marginal $P(\text{crash}) = 4.5\%$
- 210 of the 1,200 days had measurable rainfall
- Of those 210 rainy days, 27 had a crash

Question: Compute $P(\text{crash} \mid \text{rain})$. Is the marginal 4.5% a reasonable number to use when deciding whether to post a wet-weather speed advisory?

✓ Quick Check — Answer

$$P(\text{crash} \mid \text{rain}) = \frac{27}{210} = \mathbf{12.9\%}$$

Compare to the marginal: $P(\text{crash}) = 4.5\%$. Conditioning on rain nearly **triples** the crash probability.

⚠ A wet-weather signage or speed-advisory program justified using the 4.5% marginal rate would badly understate the actual risk on the days it's meant to address. The same conditioning principle that governs HVAC sizing and generator sizing governs traffic-safety countermeasures.

Predict Before You Compute

The in-class exercise uses this conditioning formula:

$$P(\text{Electricity} > 100 \text{ kWh} \mid \text{BusinessHours} = \text{Yes})$$

Marginal $P(\text{Electricity} > 100) = \mathbf{39.3\%}$

Your prediction: What do you think $P(\text{Electricity} > 100 \mid \text{BusinessHours} = \text{Yes})$ is?

- **A)** About 39% — business hours probably doesn't matter much
- **B)** Around 55–65%
- **C)** Around 75–85%
- **D)** Above 90%

✓ The Answer — $P(\text{Electricity} > 100 \mid \text{BusinessHours}) \approx 96\%$

The marginal 39.3% is a **weighted average** of two very different conditional probabilities:

Condition	$P(\text{Electricity} > 100 \text{ kWh})$	Hours
Business hours (Mon–Fri 8am–6pm)	$\approx 96\%$	2,600 hrs
Non-business hours	$\approx 16\%$	6,160 hrs
All hours (marginal)	39.3%	8,760 hrs

⚠ Sizing a backup generator for the 39.3% marginal probability means it will fail to cover business-hours demand 60% of the time — exactly when the building is occupied and the stakes are highest.

Predict — Independence

Definition: A and B are **statistically independent** if $P(A \mid B) = P(A)$ — knowing B happened tells you nothing new about A.

Equivalent test: If independent: $P(A \cap B) = P(A) \times P(B)$

It is the **multiplication rule** $P(A \cap B) = P(A \mid B) \cdot P(B)$ with the conditional replaced by plain $P(A)$ — the same rule, with a shortcut independence licenses.

Scenario: Two **stormwater detention basins** (ponds that hold storm runoff and release it slowly) serve adjacent sub-watersheds in the same city. **Overtopping** = the pond fills past its outlet and spills. From 10 years of storm records:

- $P(\text{Basin A overtops}) = 5\%$
- $P(\text{Basin B overtops}) = 5\%$

What is $P(\text{both basins overtop simultaneously})$, if they are truly independent?

✓ Independence in Practice — When It Fails

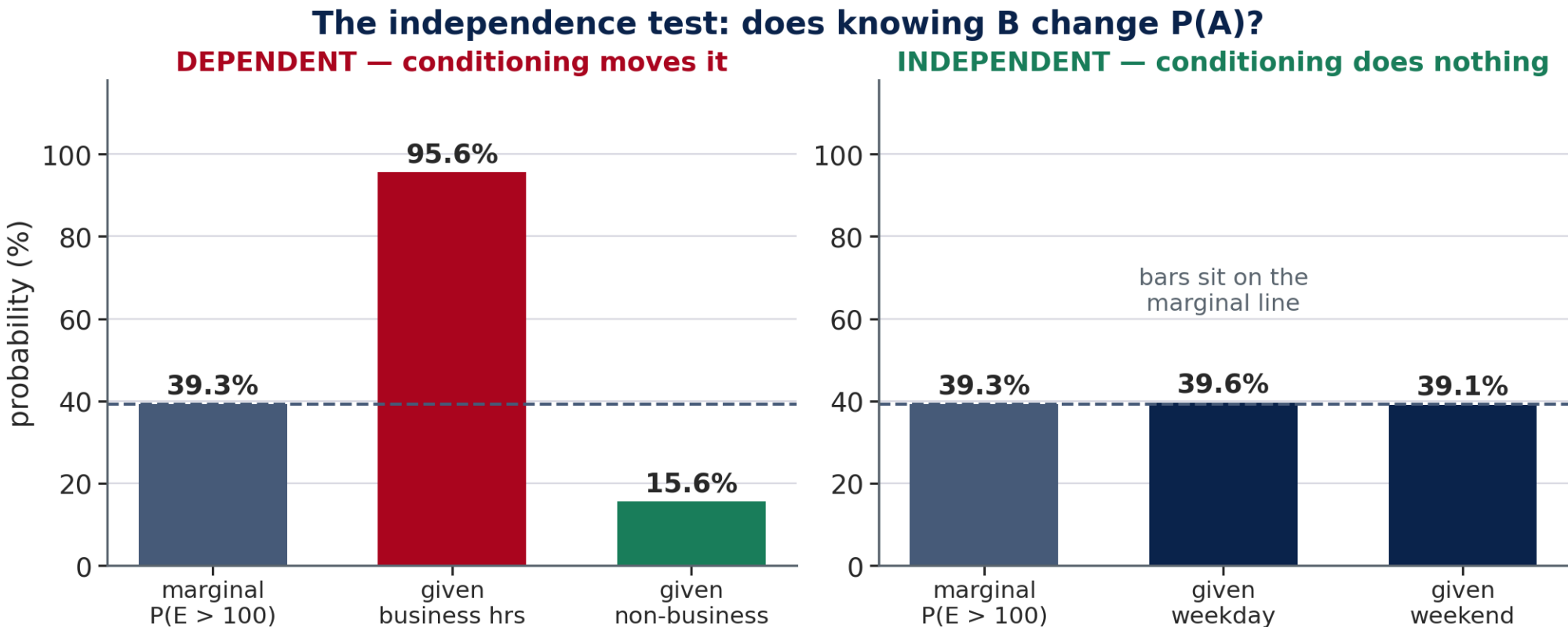
Observed over 10 years of storm events:

	Value
P(Basin A overtops)	5%
P(Basin B overtops)	5%
If independent: 0.05×0.05	0.25%
Actual observed	4.1%

Actual rate is 16× higher than independence predicts.

⚠ Both basins receive the **same storm**. Physically separate systems that share a common driver — the same storm, the same power grid, the same seismic event — are **not** independent. Assuming independence dramatically underestimates simultaneous failure risk.

The Independence Test, Drawn



Part 4

Bayes' Theorem

The Problem: You Can Only Go One Way

Counting rows gets you $P(A \mid B) = \text{Count}(A \text{ AND } B) / \text{Count}(B)$ — that is Part 3.

But some engineering problems give you $P(A \mid B)$ and ask for $P(B \mid A)$. These are not the same, and you cannot get one from the other without more information.

HVAC fault detection scenario:

You know how good the fault detection algorithm is:

- $P(\text{Alarm} \mid \text{Fault}) = \text{true positive rate}$ — alarm fires when the fault is real (correct detection)
- $P(\text{Alarm} \mid \text{No Fault}) = \text{false positive rate}$ — alarm fires when no fault exists (false alarm)

But what the maintenance team needs is the **reverse**:

- **$P(\text{Fault} \mid \text{Alarm})$** — given that the alarm fired, is there really a fault?

The Same Table, Read Two Ways

Same 1,000 hours, same 45 — a different denominator

Across the row — what you are given

	Alarm	No alarm
Fault	45	5
No fault	143	807

$P(\text{Alarm} \mid \text{Fault}) = 45 / 50 = 90\%$

Down the column — what you need

	Alarm	No alarm
Fault	45	5
No fault	143	807

$P(\text{Fault} \mid \text{Alarm}) = 45 / 188 = 24\%$

Bayes' Theorem

$$P(B | A) = \frac{P(A | B) \cdot P(B)}{P(A)}$$

Read as: "updated probability of B = (how likely the evidence is when B is true) × (prior chance of B), divided by the overall chance of seeing the evidence."

The three ingredients — in the order they happen:

Term	Name	Meaning
P(B)	Prior	What you believed before you looked — how common the condition is
P(A B)	Likelihood	What the evidence would look like if that belief were true
P(B A)	Posterior	What you believe now, with the evidence in hand

💡 Read top to bottom and the table is a story: **belief, evidence, revised belief**. The likelihood is a fact about your *instrument*; the posterior answers the question you actually *asked*. Bayes' theorem is the rule for how far the evidence is allowed to move you — and it is rarely all the way.

Bayes' Theorem — The Denominator

The denominator $P(A)$ in Bayes' theorem expands using the **total probability rule**:

$$P(A) = P(A \mid B) \cdot P(B) + P(A \mid B^c) \cdot P(B^c)$$

Read as: "overall chance of evidence A = (chance of A when B is true × how common B is) + (chance of A when B is false × how common 'not B' is)."

Why there are two terms. Evidence can arrive by two routes, and $P(A)$ counts both:

Route to the evidence A	Probability
B is true, and A follows	$P(A \mid B) \cdot P(B)$
B is false, and A happens anyway	$P(A \mid \text{not } B) \cdot P(\text{not } B)$

The second route is the one people forget. When B is rare, $P(\text{not } B)$ is close to 1, so even a small false-positive rate can make that term the **larger** of the two.

⚠ Drop the second term and $P(B \mid A)$ comes out far too high — a posterior close to your likelihood is the symptom.

HVAC Fault Detection — Setting Up the Problem

A building automation system monitors a Tucson office building's chiller. Based on maintenance records and algorithm documentation:

What we know	Value
$P(\text{Fault})$ — faults occur in 5% of summer operating hours	0.05
$P(\text{Alarm} \mid \text{Fault})$ — algorithm catches 90% of real faults	0.90
$P(\text{Alarm} \mid \text{No Fault})$ — 15% false positive rate	0.15

What the maintenance team needs: $P(\text{Fault} \mid \text{Alarm}) = ?$

Why it matters: every alarm dispatches a technician. If most alarms turn out to be noise, the team learns to ignore them — including the real one.

Before computing — what is your intuition? The algorithm is 90% accurate. Does that mean 90% of alarms are real faults?

Bayes' Theorem — Computing $P(\text{Fault} \mid \text{Alarm})$

Step 1 — Total probability of an alarm:

$$\begin{aligned} P(\text{Alarm}) &= P(\text{Alarm} \mid \text{Fault}) \cdot P(\text{Fault}) + P(\text{Alarm} \mid \text{No Fault}) \cdot P(\text{No Fault}) \\ &= 0.90 \times 0.05 + 0.15 \times 0.95 = 0.045 + 0.1425 = \mathbf{0.1875} \end{aligned}$$

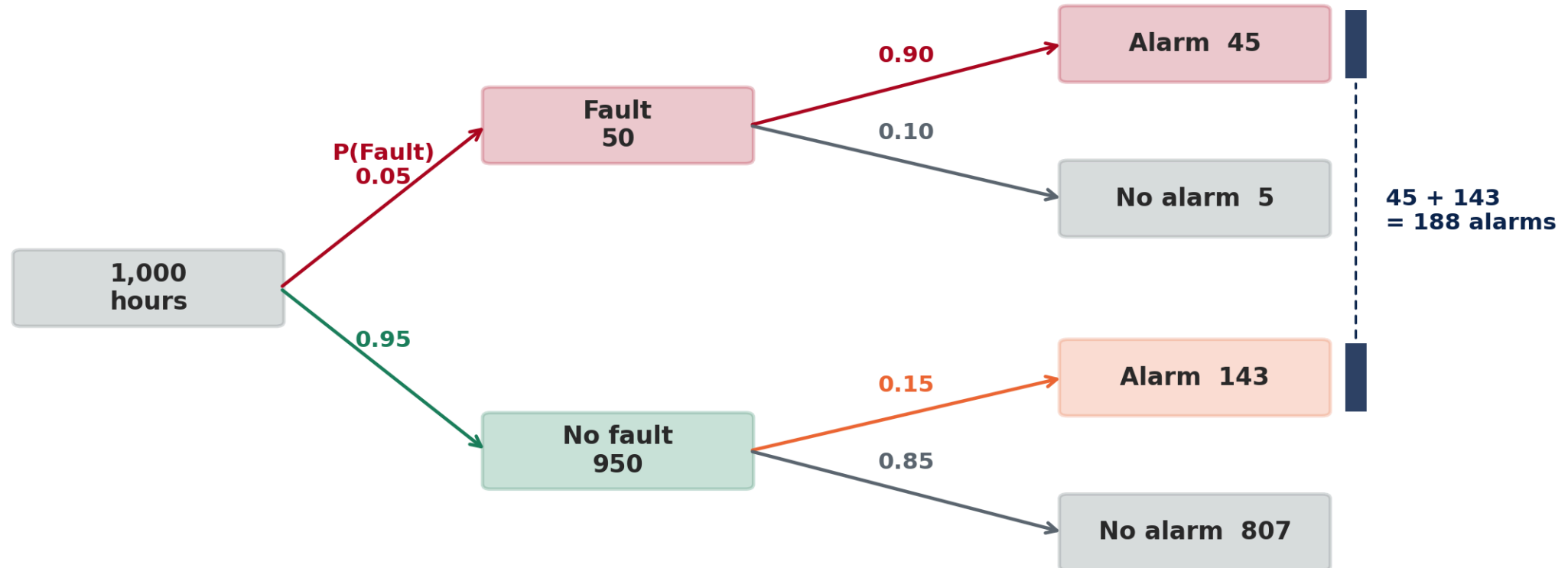
Step 2 — Apply Bayes:

$$P(\text{Fault} \mid \text{Alarm}) = \frac{P(\text{Alarm} \mid \text{Fault}) \cdot P(\text{Fault})}{P(\text{Alarm})} = \frac{0.045}{0.1875} = \mathbf{0.24}$$

Only **24% of alarms** correspond to real faults — even though the algorithm detects 90% of faults when they exist.

The Same Calculation as a Tree

Only the two 'Alarm' leaves are still in play



$$P(\text{Fault} \mid \text{Alarm}) = 45 / (45 + 143) = 24\%$$

Verify With a Frequency Table

Here is the same calculation using concrete counts — imagining 1,000 summer operating hours makes the fractions tangible.

Imagine 1,000 summer operating hours:

	Alarm fires	No alarm	Row total
Fault present	45	5	50 (5% of 1,000)
No fault	143	807	950
Column total	188	812	1,000

$$P(\text{Fault} \mid \text{Alarm}) = \frac{45}{188} = 23.9\% \approx 24\% \checkmark$$

The algorithm fires **188 alarms per 1,000 hours**. Only **45 are real** — **76% of alarms are noise**.

True Positive, False Positive – Naming the Four Cells

Bayes' theorem turns the first row into the third

1,000 summer operating hours

	Alarm fires	No alarm
Fault present	TRUE POSITIVE 45	FALSE NEGATIVE (missed fault) 5
No fault	FALSE POSITIVE (false alarm) 143	TRUE NEGATIVE 807

The same 45, three denominators

Detection rate P(Alarm Fault)	45 / 50 = 90%
False positive rate P(Alarm No fault)	143 / 950 = 15%
Precision — the one you act on P(Fault Alarm)	45 / 188 = 24%

Why the Result Surprises Everyone — The Base Rate

The 90% detection rate sounds impressive. Why is only 24% of alarms real?

Because faults are rare (5% base rate), so false positives dominate:

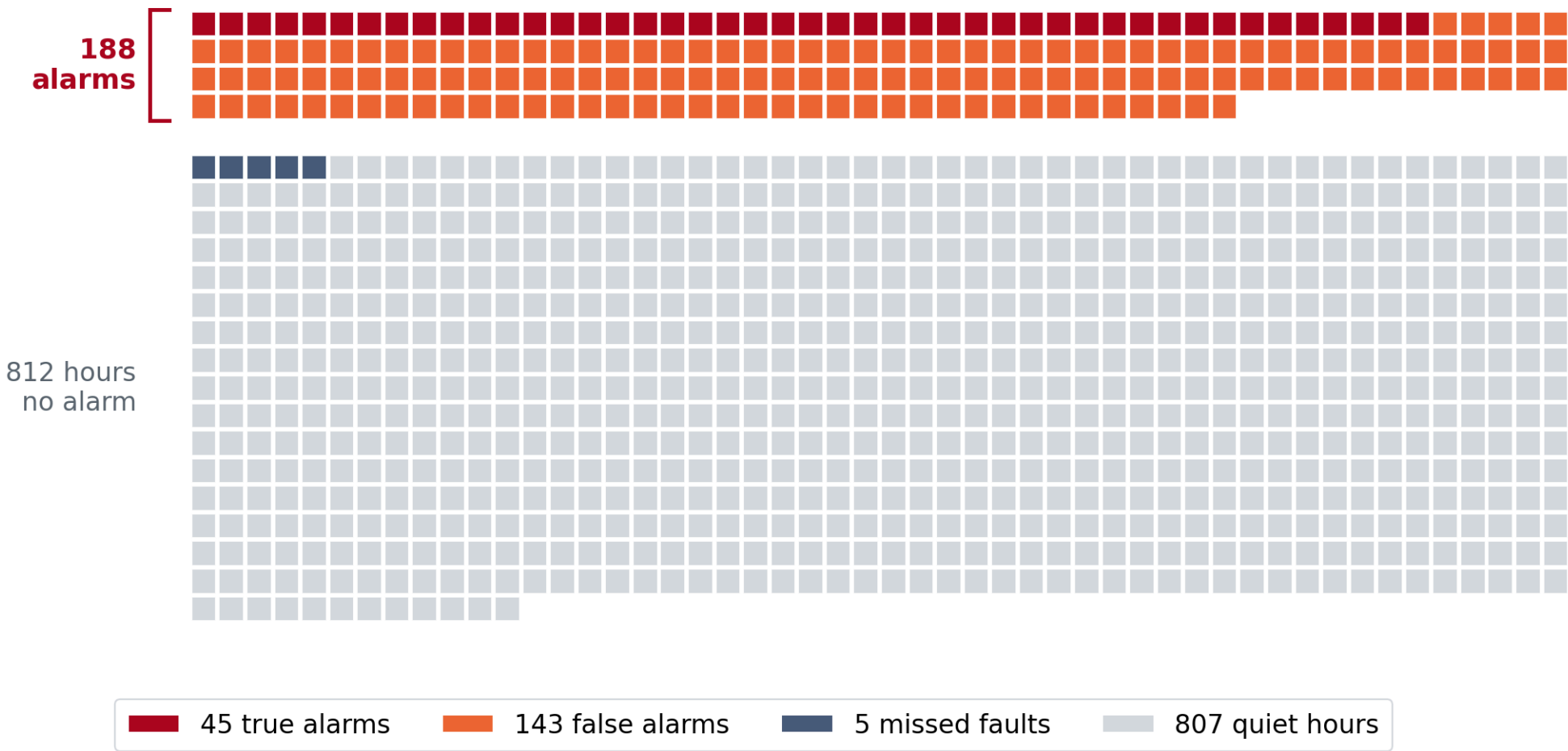
- 950 fault-free hours $\times$ 15% false positive rate = **143 false alarms**
- 50 faulty hours $\times$ 90% detection = **45 true alarms**

The false alarms outnumber the real ones 3-to-1 simply because there are 19 $\times$ more fault-free hours than faulty ones.

 **The base rate effect:** Even a highly accurate test produces mostly false positives when the condition it tests for is rare. This is why medical screening tests, structural inspection protocols, and building fault detection systems all report precision (what fraction of positives are true) alongside sensitivity (what fraction of true positives are detected). Bayes' theorem is the bridge between the two.

One Thousand Hours, One Square Each

1,000 summer hours · a 90%-accurate detector · 76% of its alarms are noise



Engineering Response — What Does 24% Mean?

$P(\text{Fault} \mid \text{Alarm}) = 24\%$ is not negligible — and not actionable as a confirmed fault.

The tiered-response protocol that Bayes justifies:

Posterior probability	Recommended response
$P(\text{Fault} \mid \text{Alarm}) < 10\%$	Log, do not dispatch
10–40%	Send technician to inspect (not repair)
40–70%	Dispatch with parts; inspect before installing
$> 70\%$	Treat as confirmed fault; repair on arrival

At 24%, the right call is **inspection, not emergency repair**. This is how modern CMMS (computerized maintenance management systems) use Bayesian fault scoring to prioritize work orders — not by binary alarm/no-alarm, but by posterior probability.

Conditional Probability — Common Mistakes

⚠ $P(A | B)$ and $P(B | A)$ use the same joint count in the numerator — they differ only in the denominator. Swapping numerator and denominator is the single most common error when working with conditional probabilities.

Quick Check — Apply Bayes Yourself

A structural health monitoring sensor is installed on a bridge deck joint. Based on inspection records and sensor specifications:

What we know	Value
$P(\text{Defect})$ — 2% of monitored joints have active cracks	0.02
$P(\text{Alert} \mid \text{Defect})$ — sensor detects 80% of real defects	0.80
$P(\text{Alert} \mid \text{No Defect})$ — 5% false positive rate	0.05

Compute: $P(\text{Defect} \mid \text{Alert}) = ?$

Apply the two-step Bayes calculation. Work individually for 3 minutes, then compare with your neighbor.

Quick Check — Answer

Step 1 — Total probability of an alert:

$$P(\text{Alert}) = 0.80 \times 0.02 + 0.05 \times 0.98 = 0.016 + 0.049 = \mathbf{0.065}$$

Step 2 — Bayes:

$$P(\text{Defect} \mid \text{Alert}) = \frac{0.016}{0.065} \approx \mathbf{24.6\%}$$

Only **1 in 4 alerts** corresponds to a real defect — even with an 80% detection rate.

Same result as the HVAC example (24%) — when the condition is rare, false positives from the majority dominate regardless of domain.

Part 5

This Week's In-Class Exercise

What You'll Do — Section A

Dataset: `Week3_Tucson_Office_Energy_2023.csv` — 8,760 hourly rows, 12 columns

Section A — Marginal Probabilities

- A1–A2: Count hours with any cooling load and with high cooling demand (> 50 kWh)
- A3: Marginal $P(\text{Cooling} > 50)$ — your baseline for all comparisons
- A4–A5: High total electricity counts and marginal $P(\text{Electricity} > 100)$
- A6–A7: Count summer hours and weekday hours
- A8: Average July electricity — first hint of what conditioning reveals

What You'll Do — Section B

Section B — Conditional Probabilities and Independence

- B1–B2: $P(\text{Cooling} > 50 \mid \text{July})$ — your first full conditional probability
- B3: $P(\text{Cooling} > 50 \mid \text{January})$ — the winter baseline
- B4: Independence test for Cooling and Month
- B5–B6: $P(\text{Electricity} > 100 \mid \text{Weekday})$
- B7–B10: Weekend and business-hours conditioning
- B11–B13: Pivot table — $P(\text{Cooling} > 50 \mid \text{Season})$, peak month, mean under peak conditions
- B14: Reverse conditioning — $P(\text{Weekday} \mid \text{July})$ — sets up Bayes' theorem

Three Deliverables This Week

 **No Monday lecture this week.** Labor Day cancels Mon Sep 7 — this lecture is posted as a **recording**. Wednesday and Friday run as normal.

	What it covers	When
Weekly Quiz	15 questions on this lecture and the Tutorial	Closes 8:00 AM Tuesday, Sep 15
In-Class Exercise	Sections A and B, the charts, and the memo	Work Wed + Fri · due Wed 9:00 AM next week
Homework	A separate take-home problem set	On your own · due Wed 9:00 AM next week

Office hours run as normal.

 Three submissions, three D2L Assignments folders. The quiz closes 8:00 AM Tuesday, Sep 15; the exercise and the homework are both due Wednesday 9:00 AM — no late work is accepted.

Looking Ahead — Week 4 Preview

At the end of this week's in-class exercise you will see this question:

"If extreme-cooling hours occur at an average rate of $\lambda = 4$ per week during July, what is the probability a randomly chosen July week has exactly 6 extreme-cooling hours?"

The question to sit with:

- Is cooling demand exactly 4 per week every week, or does it vary?
- What statistical distribution models the **count of rare events** that occur at a known average rate?

💡 Week 4 is the **Poisson distribution** — the natural model for counts of events (equipment faults, monsoon rain days, building occupancy spikes) that occur at a roughly constant average rate. It builds directly on the probabilities you compute this week.

Summary — Week 3

Conditioning is not a technicality — it is the difference between the right answer and a dangerous approximation

Today's concepts:

- **Classical probability** — complement, addition rule, independence
- **Conditional probability** — $P(A \mid B) = \text{joint count} / \text{group count}$; restricts the sample space
- **Independence test** — compare joint probability to product of marginals
- **Bayes' theorem** — reverse the conditional; update a prior with new evidence
- **Confusion matrix** — true/false positive and negative; detection rate vs. precision

The core insight: Marginal probabilities are weighted averages. Conditioning on context (season, occupancy, fault state) reveals the structure that design decisions require.

Key Formulas – Probability Rules

Concept	Formula
Complement	$P(A^c) = 1 - P(A)$
Addition rule	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Conditional probability	$P(A B) = P(A \cap B) / P(B)$
Independence test	$P(A \cap B) = P(A) \times P(B)$ if independent
Total probability	$P(A) = P(A B) \cdot P(B) + P(A B^c) \cdot P(B^c)$

Key Formulas — Bayes' Theorem & Reference

Concept	Formula
Bayes' theorem	$P(B A) = P(A B) \cdot P(B) / P(A)$